

Steering control for control-affine systems via nonlinear controllability Gramian

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(Joint work with ShiNung Ching)

Washington University in St. Louis – McKelvey School of Engineering
Association Science pour l'Afrique (ASA) Seminar

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Outline

- 1 Control systems: Examples and the controllability problem
- 2 Control-affine systems: Definition and frameworks
- 3 Linear time-invariant systems
- 4 Control-affine systems: Steering control and necessary condition for optimality
- 5 Applications and Numerical illustrations
- 6 Takeaways and Perspectives



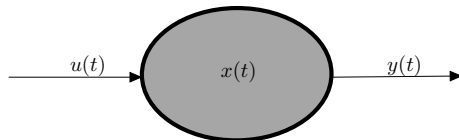
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Notion of control systems

A control system in this talk is a **dynamical system** on which one can act by using suitable **controls**.



Mathematically, it often takes the form

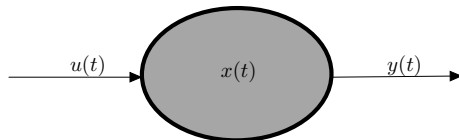
$$\dot{x}(t) = f(t, x(t), u(t)), \quad y(t) = g(t, x(t), u(t)). \quad (\Sigma)$$

- $x(\cdot)$ is the **state** of the system and $u(\cdot)$ is the **control**;
- $y(\cdot)$ is the **output**, i.e., represents what we **observe/measure** from the system;
- A solution $t \in [t_0, T] \mapsto (x(t), u(t))$ of $\dot{x}(t) = f(t, x(t), u(t))$ is called a **trajectory** of (Σ) ;
- $\Omega \subseteq \mathbb{R}^d$ is the **state space** and $\mathcal{U} \subseteq L^\infty((t_0, T); \mathbb{R}^k)$ is the set of **admissible controls**; $k \leq d$.



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Control-affine systems: reachable set & talk's focus

$$(\Sigma) : \begin{cases} \dot{x}(t) = N(x(t)) + B(x(t))u(t) \\ x(0) = x^0 \in \Omega \end{cases}$$

Reachability

For $x^0 \in \Omega$, we say that $x^1 \in \Omega$ is reachable on $[0, T]$ from x^0 if $\exists u \in L^\infty((0, T); \mathbb{R}^k)$ s.t. the solution $x_u(\cdot)$ of (Σ) satisfies $x_u(T) = x^1$.

Reachable set: formal definition

The reachable set is defined by

$$\mathcal{R}_\Sigma(T, x^0) := \left\{ x_u(T) : \begin{array}{l} x_u(\cdot) \text{ solution of } (\Sigma), \\ x(0) = x^0 \end{array} \right\}.$$



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Complete controllability

(Σ) is said to be completely controllable (or controllable) on $[0, T]$ for some $T > 0$ if $\mathcal{R}_\Sigma(T, x^0) = \Omega$ for all $x^0 \in \Omega$.

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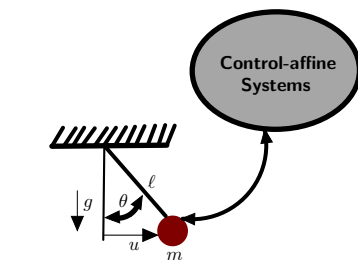
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Objective of the talk

- **Control analysis:** characterize $\mathcal{R}_\Sigma(T, x^0)$ or provide a *quantitative subset*;
- **Control synthesis:** given $x^1 \in \mathcal{R}_\Sigma(T, x^0)$, synthesize a steering control from x^0 to x^1 ;
- **Minimum-energy control:** among steering controls from x^0 to $x^1 \in \mathcal{R}_\Sigma(T, x^0)$, find one with minimum L^2 -norm.



Examples of control-affine systems

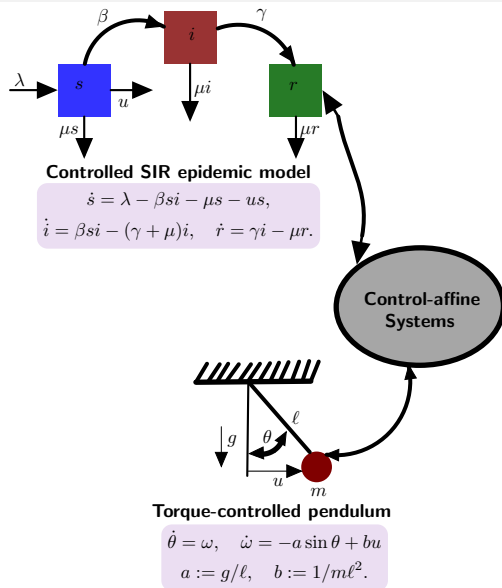


Torque-controlled pendulum

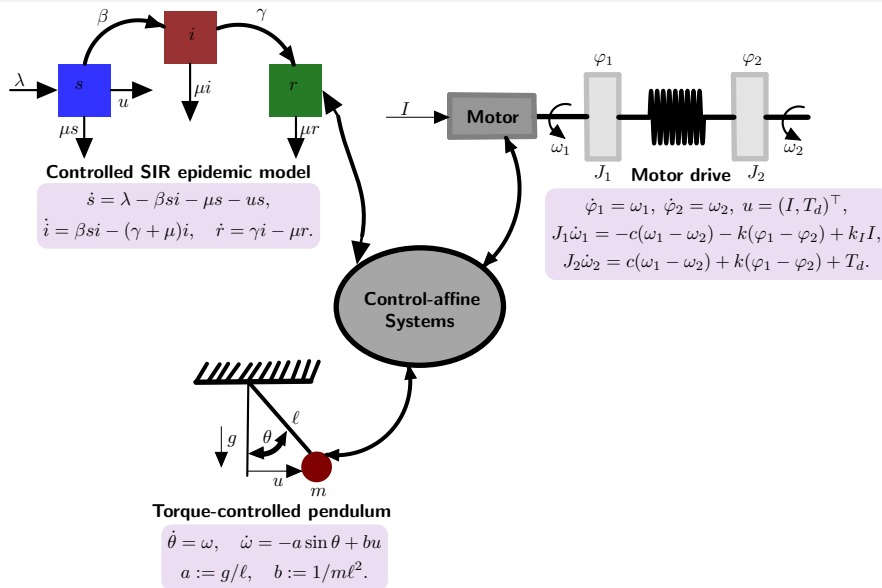
$$\begin{aligned}\dot{\theta} &= \omega, & \dot{\omega} &= -a \sin \theta + bu \\ a &:= g/\ell, & b &:= 1/m\ell^2.\end{aligned}$$



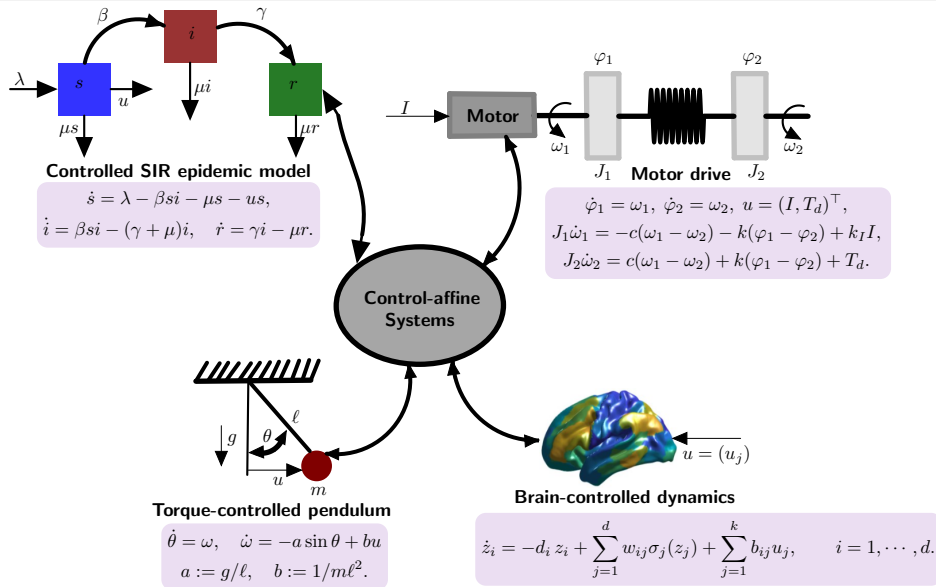
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Linear time-invariant systems [Kalman, early 60]

$$(\Sigma) : \begin{cases} \dot{x}(t) = Ax(t) + Bu(t) \\ x(0) = x^0 \in \mathbb{R}^d \end{cases}$$

Control analysis

- $A \in \mathbb{R}^{d \times d}$, $B \in \mathbb{R}^{d \times k}$, $x(\cdot) \in \mathbb{R}^d$, and $u(\cdot) \in \mathbb{R}^k$;
- $\forall x^0 \in \mathbb{R}^d$, $\mathcal{R}_\Sigma(T, x^0) = e^{TA}x^0 + \text{Range}(W_c)$
 where $W_c := \int_0^T e^{tA}BB^\top e^{tA^\top} dt$;
- $\mathcal{R}_\Sigma(T, x^0) = \mathbb{R}^d$ if and only if W_c is invertible.



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Control synthesis & Minimum-energy control

- The control $u(t) = B^\top e^{(T-t)A^\top} W_c^{-1} (x^1 - e^{TA}x^0)$ steers (Σ) from x^0 to $x^1 \in \mathcal{R}_\Sigma(T, x^0)$ in time T ;
- u is the minimum-energy control among all controls steering any $x^0 \in \mathbb{R}^d$ to $x^1 \in \mathcal{R}_\Sigma(T, x^0)$.



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- Are based on the variation of constants formula: $\forall t \in [0, T]$,
$$x(t) = e^{tA}x^0 + \int_0^t e^{(t-s)A}Bu(s) ds;$$
- Uniform w.r.t. $T > 0$ and $x^0 \in \mathbb{R}^d$: If $\mathcal{R}_\Sigma(T_0, 0) = \mathbb{R}^d$ for some $T_0 > 0$, then $\mathcal{R}_\Sigma(T, x^0) = \mathbb{R}^d$, $\forall T > 0$ and $\forall x^0 \in \mathbb{R}^d$;
- Does not account for nonlinearities in $x(\cdot)$ and/or $u(\cdot)$.



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Frameworks, flow maps of the drift

$$(\Sigma) : \begin{cases} \dot{x}(t) = N(x(t)) + B(x(t))u(t) \\ x(0) = x^0 \in \mathbb{R}^d \end{cases}$$

Assumption on the drift term

- $N \in C^2(\mathbb{R}^d)$ is globally Lipschitz and $\|DN(x)\| \leq \Lambda_1, \forall x \in \mathbb{R}^d$;
- $\|D^2N(x)\| \leq \Lambda_2, \forall x \in \mathbb{R}^d$.

Assumption on the input matrix & control

- $B \in L^\infty(\mathbb{R}^d; \mathbb{R}^{d \times k}), x \mapsto B(x)$ is $C^1(\mathbb{R}^d)$ and globally Lipschitz;
- $u \in L^\infty((0, T); \mathbb{R}^k)$.



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- $(\phi_t)_{t \in \mathbb{R}}$ flow of N : $\partial_t \phi_t(x) = N(\phi_t(x)), \phi_0(x) = x$;
- The Jacobian matrix $D\phi_t(x)$ satisfies:
 $\partial_t D\phi_t(x) = DN(\phi_t(x))D\phi_t(x), D\phi_0(x) = \text{Id}.$

$\forall \alpha_t \in C^1(\mathbb{R}_+; \mathbb{R}), \alpha_0 \geq 0, \forall \beta_t \in C^0(\mathbb{R}_+; \mathbb{R}^d), \forall t \geq 0$:

- $\|D\phi_{\alpha_t}(\beta_t)\| \leq e^{\Lambda_1 \alpha_0} e^{\Lambda_1 \int_0^t |\dot{\alpha}(\tau)| d\tau}$;
- $\|D^2\phi_{\alpha_t}(\beta_t)\| \leq \Lambda_2 e^{2\Lambda_1 \alpha_0} e^{\Lambda_1 \int_0^t |\dot{\alpha}(\tau)| d\tau} (\alpha_0 + e^{\Lambda_1 \int_0^t |\dot{\alpha}(\tau)| d\tau} \int_0^t |\dot{\alpha}(\tau)| d\tau).$



Solution representation & Gramian maps

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Theorem 1 [T, Ching; '25], [Agrachev, Gamkrelidze; '79]

For all $x^0 \in \mathbb{R}^d$ and $u \in L^\infty := L^\infty((0, T); \mathbb{R}^k)$ the solution $x_u(\cdot) \in C^0([0, T]; \mathbb{R}^d)$ of (Σ) expresses for $t \in [0, T]$ as

$$x_u(t) = \phi_{t-T} \left(\phi_T(x^0) + \int_0^t D\phi_{T-s}(x_u(s)) B(x_u(s)) u(s) ds \right).$$



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Definition [T, Ching; '25]

Fix $x^0 \in \mathbb{R}^d$. The **Gramian map** $\mathcal{N} : L^\infty \rightarrow \mathcal{S}_d^+(\mathbb{R})$ is defined for every $u \in L^\infty$ and $x_u(\cdot)$ solution of (Σ) by:

$$\mathcal{N}(u) = \int_0^T D\phi_{T-t}(x_u(t)) B(x_u(t)) B(x_u(t))^\top D\phi_{T-t}(x_u(t))^\top dt.$$



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If $N(x) = Ax$ and $B(x) = B$, then

- $x_u(t) = e^{tA}x^0 + \int_0^t e^{(t-s)A}Bu(s) ds;$

- $\mathcal{N}(u) = W_c = \int_0^T e^{tA}BB^\top e^{tA^\top} dt.$

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A priori analysis

- ① From $x_u(t) = \phi_{t-T} \left(\phi_T(x^0) + \int_0^t D\phi_{T-s}(x_u(s)) B(x_u(s)) u(s) ds \right)$, one has $x_u(T) = x^1$, iff

$$\int_0^T D\phi_{T-t}(x_u(t)) B(x_u(t)) u(t) dt = x^1 - \phi_T(x^0). \quad (1)$$

- ② Fix $u \in L^\infty$, define the bounded linear operator $\mathcal{L}_u : L^2 := L^2((0, T); \mathbb{R}^k) \rightarrow \mathbb{R}^d$ by

$$\mathcal{L}_u v = \int_0^T D\phi_{T-t}(x_u(t)) B(x_u(t)) v(t) dt \quad v \in L^2. \quad (2)$$

- ③ If for some $u \in L^\infty$, $y := x^1 - \phi_T(x^0) \in \text{Range}(\mathcal{L}_u)$, then $\min\{\|v\|_{L^2} : \mathcal{L}_u v = y\}$ has the solution

$$v(u)(t) := (\mathcal{L}_u^\dagger y)(t) = B(x_u(t))^\top D\phi_{T-t}(x_u(t))^\top \mathcal{N}(u)^\dagger y \quad t \in [0, T]. \quad (3)$$



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Feasible coercivity class

Key remark

If one can prove for some $u \in L^\infty$ that $\mathcal{N}(u)$ is invertible and u satisfies the implicit relation

$$u(t) = B(x_u(t))^\top D\phi_{T-t}(x_u(t))^\top \mathcal{N}(u)^{-1} (x^1 - \phi_T(x^0)) \quad t \in [0, T], \quad (4)$$

then the solution $x_u(\cdot)$ of (Σ) with control $u(\cdot)$ satisfies $x_u(T) = x^1$.

Definition [T, Ching; '25]

Let $C := C(\|B\|_\infty, \Lambda_1, \Lambda_2, T, \text{ and possibly } x^0) > 0$, we define the *feasible coercivity class* by

$$\mathcal{F}(C) := \left\{ u \in L^\infty((0, T); \mathbb{R}^k) : \lambda_{\min}(\mathcal{N}(u)) \geq C^{-1} \right\}. \quad (5)$$

- *Possibility of emptiness:* For a given $C > 0$, $\mathcal{F}(C) = \emptyset$ is possible;
- *Monotonicity:* If $C \leq C'$ then $\mathcal{F}(C) \subset \mathcal{F}(C')$;
- *Topological property:* $\mathcal{F}(C)$ is closed in L^∞ since $u \mapsto \lambda_{\min}(\mathcal{N}(u))$ is continuous.



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Synthesis maps & feasible coercivity ball

Definition [T, Ching; '25]

- *Feasible coercivity class:* $\mathcal{F}(C) := \left\{ u \in L^\infty((0, T); \mathbb{R}^k) : \lambda_{\min}(\mathcal{N}(u)) \geq C^{-1} \right\};$
- *Synthesis map.* $\mathcal{S} : \mathcal{F}(C) \rightarrow L^\infty$ by

$$\mathcal{S}(u)(t) = B(x_u(t))^\top D\phi_{T-t}(x_u(t))^\top \mathcal{N}(u)^{-1} y \quad u \in \mathcal{F}(C), t \in [0, T]. \quad (6)$$

- *A priori estimate on $\mathcal{F}(C)$:*

$$\|\mathcal{S}(u)\|_\infty \leq \zeta(y) := C \|B\|_\infty e^{\Lambda_1 T} |y| \quad \forall u \in \mathcal{F}(C). \quad (7)$$

- *Feasible coercivity ball:*

$$\mathcal{F}(y) := \{u \in \mathcal{F}(C) : \|u\|_\infty \leq \zeta := \zeta(y)\}. \quad (8)$$



Fixed point of the synthesis map via Banach fixed-point theorem

Theorem 2 [T, Ching; '25]

Assume that $\mathcal{F}(C) \neq \emptyset$ and $\mathcal{S}(\mathcal{F}(y)) \subset \mathcal{F}(C)$. There exists $K > 0$, $p \in (0, 1)$ and $\gamma_n(p) \rightarrow 0$ as $n \rightarrow \infty$ such that

$$\|\mathcal{S}^n(u) - \mathcal{S}^n(v)\|_\infty \leq \varrho_n \|u - v\|_\infty, \quad \forall (u, v) \in \mathcal{F}(y)^2, \quad \forall n \geq 1, \quad (9)$$

where $\varrho_n := K^n \gamma_n(p)$. Furthermore, if there exists $n_0 \geq 1$ such that $\varrho_{n_0} < 1$, then $\mathcal{S}$ admits a unique fixed point $u_* \in \mathcal{F}(y)$, and the Picard iteration $u^{(n+1)} = \mathcal{S}(u^{(n)})$ converges to u_* for any $u^{(0)} \in \mathcal{F}(y)$.

- ① Two key assumptions: $\mathcal{F}(C) \neq \emptyset$ and self-mapping condition $\mathcal{S}(\mathcal{F}(y)) \subset \mathcal{F}(C)$;
- ② Consequence of the estimate

$$|\mathcal{S}(u)(t) - \mathcal{S}(v)(t)| \leq \mathcal{V}|u - v|(t) \quad \forall t \in [0, T], \quad \forall (u, v) \in \mathcal{F}(y)^2, \quad (10)$$

where $\mathcal{V} : L^\infty \rightarrow L^\infty$ is a Volterra-rank one linear operator, which is compact and positive on the cone of nonnegative functions in L^∞ .



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$$\|\mathcal{S}^n(u) - \mathcal{S}^n(v)\|_\infty \leq \varrho_n \|u - v\|_\infty, \quad \forall (u, v) \in \mathcal{F}(y)^2, \quad \forall n \geq 1, \quad (9)$$

where $\varrho_n := K^n \gamma_n(p)$. Furthermore, if there exists $n_0 \geq 1$ such that $\varrho_{n_0} < 1$, then $\mathcal{S}$ admits a unique fixed point $u_* \in \mathcal{F}(y)$, and the Picard iteration $u^{(n+1)} = \mathcal{S}(u^{(n)})$ converges to u_* for any $u^{(0)} \in \mathcal{F}(y)$.

- ① Two key assumptions: $\mathcal{F}(C) \neq \emptyset$ and self-mapping condition $\mathcal{S}(\mathcal{F}(y)) \subset \mathcal{F}(C)$;
- ② Consequence of the estimate

$$|\mathcal{S}(u)(t) - \mathcal{S}(v)(t)| \leq \mathcal{V}|u - v|(t) \quad \forall t \in [0, T], \quad \forall (u, v) \in \mathcal{F}(y)^2, \quad (10)$$

where $\mathcal{V} : L^\infty \rightarrow L^\infty$ is a Volterra-rank one linear operator, which is compact and positive on the cone of nonnegative functions in L^∞ .



Steering control and necessary condition for optimality

Theorem 3 [T, Ching; '25]

Let $(x^0, x^1) \in \mathbb{R}^d \times \mathbb{R}^d$ and set $y := x^1 - \phi_T(x^0)$. Assume $\mathcal{F}(C) \neq \emptyset$, $\mathcal{S}(\mathcal{F}(y)) \subset \mathcal{F}(C)$, and that the iterate-contraction criterion of Theorem 2 is satisfied. Let $\bar{u} \in \mathcal{F}(y)$ be the fixed point of $\mathcal{S}$ given by

$$\bar{u}(t) = B(x_{\bar{u}}(t))^{\top} D\phi_{T-t}(x_{\bar{u}}(t))^{\top} \mathcal{N}(\bar{u})^{-1}y. \quad (11)$$

Then $\bar{u}$ steers (Σ) from x^0 to x^1 on $[0, T]$. Furthermore, one has the energy identity

$$\|\bar{u}\|_{L^2}^2 = y^{\top} \mathcal{N}(\bar{u})^{-1}y. \quad (12)$$

Moreover, if $\bar{u}$ is a local minimum of $\frac{1}{2}\|u\|_{L^2}^2$ over $\mathfrak{F} := \{u \in \mathcal{F}(y) : F(u) := \mathcal{L}_u u - y = 0\}$, then

$$\langle [\mathcal{L}_{\bar{u}} DF(\bar{u})^*]^{-1}y, DF(\bar{u})h \rangle_{\mathbb{R}^d} = 0 \quad \forall h \in \ker \mathcal{L}_{\bar{u}}. \quad (13)$$

- Fundamentally a synthesis result: given the fixed point $\bar{u} \in \mathcal{F}(y)$ of $\mathcal{S}$, it provides an explicit representation of a control that steers x^0 to x^1 ;
- Reachability depends on the structure of $\mathcal{F}(C)$ and on whether the self-mapping $\mathcal{S}(\mathcal{F}(y)) \subset \mathcal{F}(C)$ is satisfied.



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Outline

- 1 Control systems: Examples and the controllability problem
- 2 Control-affine systems: Definition and frameworks
- 3 Linear time-invariant systems
- 4 Control-affine systems: Steering control and necessary condition for optimality
- 5 Applications and Numerical illustrations**
- 6 Takeaways and Perspectives



Fully-actuated control-affine system with an invertible input matrix

Proposition 1 [T, Ching; '25]

Suppose $d = k$ and there exists $b > 0$ such that

$$\sigma_{\min}(B(x)) \geq b, \quad \forall x \in \mathbb{R}^d. \quad (14)$$

Then $\mathcal{N}(u) := \mathcal{N}(u, x^0)$ is uniformly coercive,

$$z^\top \mathcal{N}(u) z \geq \frac{1 - e^{-2\Lambda_1 T}}{2\Lambda_1} b^2 |z|^2, \quad \forall (z, x^0, u) \in \mathbb{R}^d \times \mathbb{R}^d \times L^\infty. \quad (15)$$

- 1 Our steering control: $\bar{u}(t) = B(x_{\bar{u}}(t))^\top D\phi_{T-t}(x_{\bar{u}}(t))^\top \mathcal{N}(\bar{u})^{-1}y$ steers (Σ) from any $x^0 \in \mathbb{R}^d$ to any $x^1 \in \mathbb{R}^d$;
- 2 Feedback linearization control: $u_{fl}(t) = B(x(t))^{-1} \left(\frac{x^1 - x^0}{T} - N(x(t)) \right)$ steers (Σ) from any $x^0 \in \mathbb{R}^d$ to any $x^1 \in \mathbb{R}^d$.



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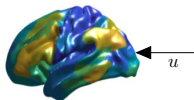
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Fully-actuated 100D continuous-time Hopfield-type RNN

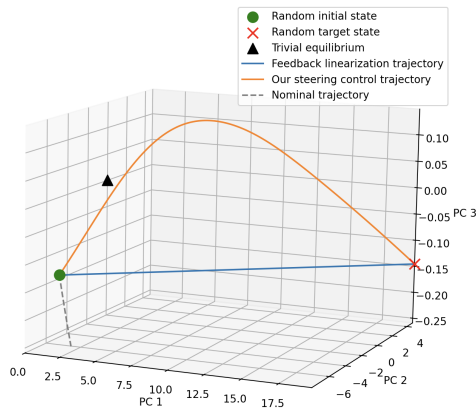
100D MINDy models [Chen *et al.*, '25]



$$\dot{x}_i = -d_i x_i + \sum_{j=1}^{100} w_{ij} \sigma_j(x_j) + \sum_{j=1}^{100} \delta_{ij} u_j, \quad i = 1, \dots, 100.$$

- Here, δ_{ij} is Kronecker's symbol;
- $d_i > 0$, and $w_{ij} \in \mathbb{R}$ are fitted on individual Brain dynamics using fMRI;
- $\sigma_j(x_j) = \frac{\sqrt{\alpha_j^2 + (bx_j + 0.5)} - \sqrt{\alpha_j^2 + (bx_j - 0.5)}}{2}$, where $\alpha_j > 0$ and $b > 0$ are also fitted.

Controlled Trajectories in PCA space



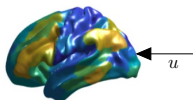
$$T = 1, \|\bar{u}\|_2 = 23.057, \|\bar{u}\|_\infty = 4.67, |x_{\bar{u}}(T) - x^1| = 10^{-11};$$

$$\|u_{fl}\|_2 = 23.049, \|u_{fl}\|_\infty = 4.61, |x_{u_{fl}}(T) - x^1| = 10^{-13}.$$



Fully-actuated 3D continuous-time Hopfield-type RNN

A 3D toy example



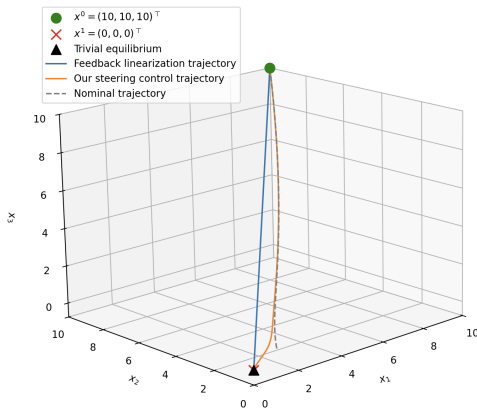
$$\dot{x}_i = -d_i x_i + \sum_{j=1}^3 w_{ij} \sigma_j(x_j) + \sum_{j=1}^3 \delta_{ij} u_j, \quad i = 1, 2, 3.$$

• Here, $d_1 = 1.25$, $d_2 = 1.5$, $d_3 = 1$;

• $\sigma_j(x_j) := (1 + e^{-x_j})^{-1}$, and

$$(w_{ij}) = \begin{pmatrix} 3 & 1 & -0.5 \\ 2 & 1 & 0.5 \\ 0 & -1.5 & 1.25 \end{pmatrix}.$$

Controlled Trajectories



$$T = 5, \quad \|\bar{u}\|_2 = 4.60, \quad \|\bar{u}\|_\infty = 4.33, \quad |x_{\bar{u}}(T) - x^1| = 10^{-11}, \\ \|u_{fl}\|_2 = 16.12, \quad \|u_{fl}\|_\infty = 9.50, \quad |x_{u_{fl}}(T) - x^1| = 10^{-14}.$$



General sufficient conditions for the existence of our steering control

Proposition 2 [T, Ching; '25]

Let $x^0 \in \mathbb{R}^d$. Assume that $\exists u^b := u^b(x^0) \in L^\infty$ s.t. $\mathcal{N}(u^b) := \mathcal{N}(u^b, x^0)$ is invertible and

$$\|u^b\|_\infty < \frac{\lambda_{\min}(\mathcal{N}(u^b))}{2L_{\mathcal{N}}}. \quad (16)$$

Then, letting $C := 2/\lambda_{\min}(\mathcal{N}(u^b))$ yields $\mathcal{S}(\mathcal{F}(y)) \subset \mathcal{F}(C)$ for every $x^1 \in \mathbb{R}^d$ satisfying

$$|y| := |x^1 - \phi_T(x^0)| \leq \frac{\lambda_{\min}(\mathcal{N}(u^b))}{2\|B\|_\infty e^{\Lambda_1 T}} \left(\frac{\lambda_{\min}(\mathcal{N}(u^b))}{2L_{\mathcal{N}}} - \|u^b\|_\infty \right). \quad (17)$$

Here $L_{\mathcal{N}} > 0$ is the Lipschitz constant of the map $u \in \{u \in L^\infty : \|u\|_\infty \leq \zeta\} \mapsto \mathcal{N}(u)$.

- 1 **Our steering control:** $\bar{u}(t) = B(x_{\bar{u}}(t))^\top D\phi_{T-t}(x_{\bar{u}}(t))^\top \mathcal{N}(\bar{u})^{-1}y$ steers (Σ) from any $x^0 \in \mathbb{R}^d$ to any $x^1 \in \mathbb{R}^d$ whenever (17) is satisfied;
- 2 Other steering controls are available if (Σ) is **completely controllable**: homotopy continuation or nipolent approximation [Sussmann, Chitour, Jean, Trélat, Long, Schmoderer, Chen, Ji, Boizot, Gauthier, Prandi,...]



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A 3D driftless system: the unicycle

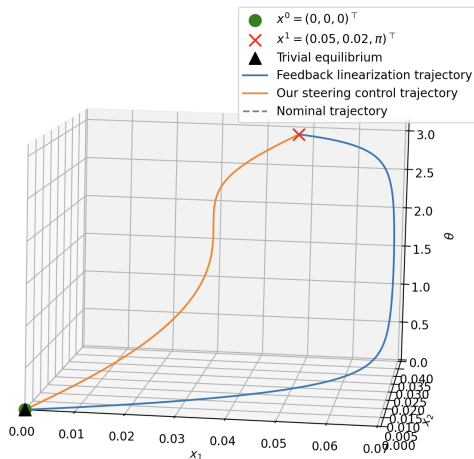
Unicycle kinematic model

$$\dot{x} = \begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{\theta} \end{pmatrix} = \begin{bmatrix} \cos \theta & 0 \\ \sin \theta & 0 \\ 0 & 1 \end{bmatrix} \begin{pmatrix} u_x \\ u_\theta \end{pmatrix} =: B(x)u$$

- The unicycle is completely controllable on $[0, T]$ for any $T > 0$;
- The unicycle admits^a a feedback linearization steering control u_{fl} from any $x^0 \in \mathbb{R}^2 \times \mathbb{S}^1$ to any $x^1 \in \mathbb{R}^2 \times \mathbb{S}^1$.

^aSiciliano B., Sciavicco L., Villani L., and Oriolo G., Robotics: Modelling, Planning and Control, Springer, 2009.

Controlled Trajectories

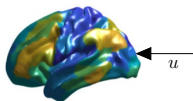


$$T = 0.15, \|\bar{u}\|_2 = 8.11, \|\bar{u}\|_\infty = 20.94, |x_{\bar{u}}(T) - x^1| = 10^{-15}, \\ \|u_{fl}\|_2 = 13.74, \|u_{fl}\|_\infty = 116.75, |x_{u_{fl}}(T) - x^1| = 10^{-15}$$



3D continuous-time Hopfield-type RNN with 2D input

A 3D toy example



$$\dot{x}_i = -d_i x_i + \sum_{j=1}^3 w_{ij} \sigma_j(x_j) + \sum_{j=1}^2 \delta_{ij} u_j, \quad i = 1, 2, 3.$$

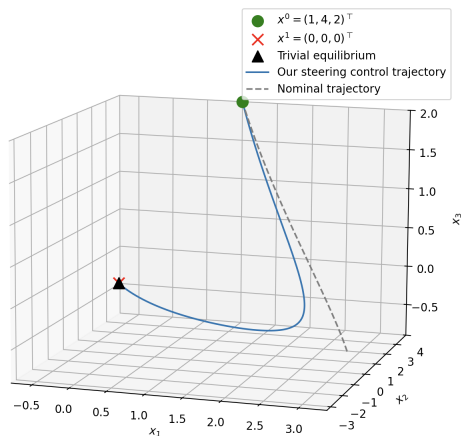
• Here, $d_1 = 1.25$, $d_2 = 1.5$, $d_3 = 1$;

• $\sigma_j(x_j) := (1 + e^{-x_j})^{-1}$, and

$$(w_{ij}) = \begin{pmatrix} 3 & 1 & -0.5 \\ 2 & 1 & 0.5 \\ 0 & -1.5 & 1.25 \end{pmatrix}.$$

Remark: We do not know *a priori* that the system is completely controllable on $[0, 5]$.

Controlled Trajectories

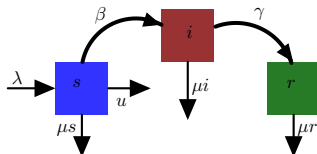


$$T = 5, \quad \|\bar{u}\|_2 = 6.57, \quad \|\bar{u}\|_\infty = 5.16, \quad |x_{\bar{u}}(T) - x^1| = 10^{-11}.$$



3D SIR model with 1D input

A 3D SIR toy example

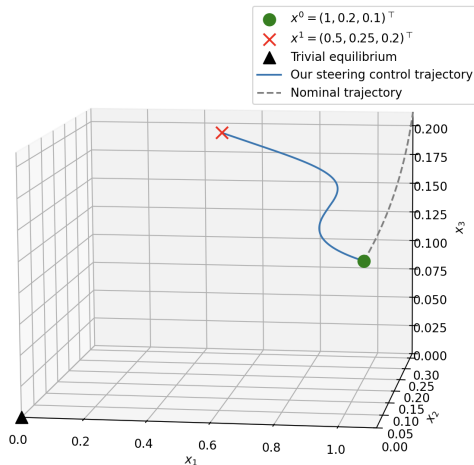


Controlled SIR epidemic model

$$\begin{aligned}\dot{s} &= \lambda - \beta si - \mu s - us, \\ \dot{i} &= \beta si - (\gamma + \mu)i, \quad \dot{r} = \gamma i - \mu r.\end{aligned}$$

For simulations, we use $\lambda = 1$, $\beta = 2$, $\mu = 0.2$ and $\gamma = 1$.

Controlled Trajectories



$$T = 0.5, \quad \|\bar{u}\|_2 = 1.91, \quad \|\bar{u}\|_\infty = 6, \quad |x_{\bar{u}}(T) - x^1| = 10^{-11}$$

Remark: We do not know *a priori* that the system is completely controllable on $[0, 0.5]$.



Planar control-affine systems with scalar inputs

Proposition 3 [T, Yu, Ching; '26]

Assume

$$\inf_{x \in \mathbb{R}^2} |\det(B, DN(x)B)| \geq \beta > 0. \quad (18)$$

Fix $x^0 \in \mathbb{R}^2$ and $U > 0$. Then there exists $\alpha_{\mathcal{N}} = \alpha_{\mathcal{N}}(\beta, |x^0|, U) > 0$ such that

$$\mathcal{N}(u) \succeq \alpha_{\mathcal{N}} \text{Id}, \quad \forall u \in L^\infty(t_0, T) \text{ with } \|u\|_\infty \leq U.$$

- 1 Proposition 3 is insightful merely when there is no control constraints *a priori*;
- 2 In this case, **our steering control** $\bar{u}(t) = B^\top D\phi_{T-t}(x_{\bar{u}}(t))^\top \mathcal{N}(\bar{u})^{-1}y$ steers (Σ) from any $x^0 \in \mathbb{R}^2$ to any $x^1 \in \mathbb{R}^2$ whenever (18) is satisfied.



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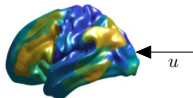
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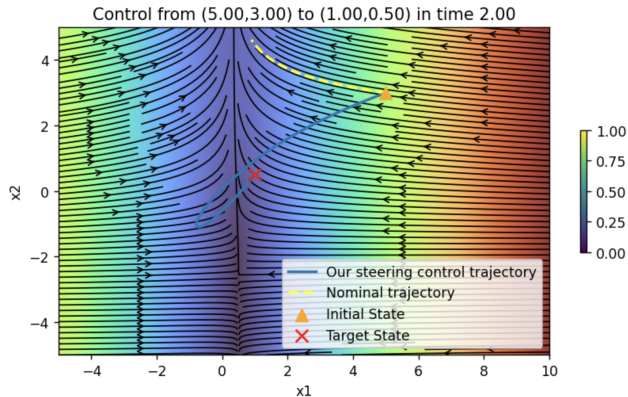
2D continuous-time Hopfield-type RNN with a scalar input

A 2D toy example



$$\dot{x}_i = -d_i x_i + \sum_{j=1}^2 w_{ij} \sigma_j(x_j) + u, \quad i = 1, 2.$$

- $d_i > 0$, $w_{ij} \in \mathbb{R}$ and $\sigma_i(\cdot)$ is a bounded sigmoid;
- The system is completely controllable on $[0, T]$ for any $T > 0$ by **[Sun, '07]**;
- For simulations: $d_1 = 1.25$, $d_2 = 0.01$,
 $\sigma_j(x_j) = \tanh x_j$, $(w_{ij}) = \begin{pmatrix} 1 & -0.15 \\ -0.15 & 1 \end{pmatrix}$.

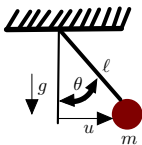


$$\|\bar{u}\|_2 = 4.85, \quad \|\bar{u}\|_\infty = 5.46, \quad |x_{\bar{u}}(T) - x^1| = 10^{-11}.$$



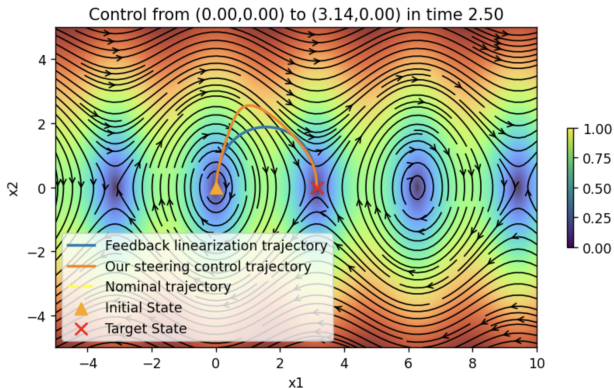
Torque-controlled pendulum (TCP)

Torque-controlled pendulum



$$\begin{aligned}\dot{\theta} &= \omega, & \dot{\omega} &= -a \sin \theta + bu \\ a &:= g/\ell, & b &:= 1/m\ell^2.\end{aligned}$$

- TCP is completely controllable on $[0, T]$ for any $T > 0$;
- $u_{fl}(t) = (v(t) + a \sin x_1(t))/b$ is a feedback linearization steering control from any $x^0 = (\theta^0, \omega^0) \in \mathbb{S}^1 \times \mathbb{R}$ to any $x^1 = (\theta^1, \omega^1) \in \mathbb{S}^1 \times \mathbb{R}$ on $[0, T]$. Here, $v(\cdot)$ is the Gramian-based control for $\dot{\theta} = \omega$, $\dot{\omega} = v$;
- For simulation: $g = 9.81$, $\ell = 4$ and $m = 1$.



$$\begin{aligned}\|\bar{u}\|_2 &= 63.68, \quad \|\bar{u}\|_\infty = 96.94, \quad |x_{\bar{u}}(T) - x^1| = 10^{-11}; \\ \|u_{fl}\|_2 &= 57.69, \quad \|u_{fl}\|_\infty = 48.25, \quad |x_{u_{fl}}(T) - x^1| = 10^{-13}.\end{aligned}$$



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Takeaways and Perspectives

Takeaways

- **Goal:** explicit steering controls for control-affine systems $\dot{x} = N(x) + B(x)u$ (endpoint control with low L^2 energy).
- **Key idea:** a *flow-conjugate* solution representation yields a nonlinear Gramian $\mathcal{N}(u)$ (LTI Gramian as a special case).
- **Synthesis:** define a map $\mathcal{S}$ whose fixed points are steering controls; Picard iterations converge under a coercivity/self-mapping regime.
- **Energy:** the fixed point satisfies $\|\bar{u}\|_{L^2}^2 = y^\top \mathcal{N}(\bar{u})^{-1} y$, and an orthogonality condition for local minimizers is provided.
- **Numerics:** validated on nonlinear benchmarks (pendulum, unicycle, SIR model, Hopfield-type networks), with clear gains vs. naive/Feedback-linearization controls.

Perspectives

- Understand when *complete controllability* can be effectively combined with the present framework to guarantee well-definiteness of our steering control.
- Develop *coercivity criteria* in higher-dimensional settings, guaranteeing well-definiteness of our steering control.
- Characterize nonlinear control-affine systems for which our steering control satisfies the *orthogonality condition*, guaranteeing a local minimizer.



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Thank You!

Questions?

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